

Which Contractions of the Octonions Give Genuine R-Flux: A Necessary and Sufficient Condition

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Abstract. Among the 56 structurally valid Inönü–Wigner contractions of the imaginary octonions sharing the Günaydin–Lüst–Malek grading shape (established in a companion note), exactly half yield the genuine, non-associative R-flux algebra, and exactly half collapse to an ordinary, fully associative Heisenberg algebra. We show, by direct and exhaustive computation, that this split is governed by a single, exact, necessary and sufficient condition: **the contraction is non-trivial if and only if the momentum subspace, not the coordinate subspace, is the one that closes into a subalgebra on its own.** The reason is mechanistic, not coincidental: the coordinate bracket $\{X, X\} \propto P$ is the defining non-associative content of R-flux, while $\{P, P\} = 0$ is already the expected, required form — so closure on the momentum side costs nothing, while closure on the coordinate side destroys the one structure the whole construction depends on.

1. Background

A companion note shows that 56 of the 140 possible $1+3+3$ gradings of $\text{Im}(\mathbb{O})$ satisfy the Weimar-Woods condition for a well-defined Inönü–Wigner contraction ($X \rightarrow \lambda X, P \rightarrow \lambda^2 P, W \rightarrow \lambda^3 W$), and that these 56 fall into exactly two orbits of 28 under the octonion automorphism group, distinguished by which of $\{X, P\}$ is itself one of the seven Fano-line subalgebras. That note reports that all 56 satisfy the Malcev identity. This is true, but incomplete on its own: Malcev-preservation does not by itself distinguish a genuine non-associative structure from a degenerate one, since every associative (Lie) algebra trivially satisfies the strictly weaker Malcev identity. This note isolates and proves the sharper, physically relevant distinction.

2. The two possible outcomes

For a valid grading with W central (grade 3), X at grade 1, P at grade 2, the contracted bracket keeps, for each pair of generators, only the component landing exactly at the sum of their grades (all lower-grade components vanish under $\lambda \rightarrow 0$; no higher-grade component is present, by the validity condition itself). The two brackets of interest:

$\{X, X\}$: grade $1 + 1 = 2 \implies$ survives iff $[X, X]_{\text{raw}}$ has a component at grade 2 (i.e. in P).

$\{P, P\}$: grade $2 + 2 = 4 \implies$ always vanishes, since no grade 4 exists in a $(1, 2, 3)$ -graded system.

So $\{P, P\} = 0$ unconditionally, for every valid grading — this is never the source of triviality. The only question is whether $\{X, X\}$ survives.

3. The exact condition

Claim. $\{X, X\} \neq 0$ in the contracted limit if and only if P (not X) is the closed Fano-line subalgebra among the two.

Mechanism. If X is the closed Fano line, then by definition $[X, X] \subseteq X$ — entirely at grade 1, with zero component at grade 2. Since only the grade-2 component of $[X, X]$ can survive contraction, the surviving part is exactly zero. If instead P is the closed Fano line, X is *not* closed, and the raw octonion structure forces $[X, X]$ to land in the remaining

available direction — which, since a W -component is already excluded by the Weimar-Woods validity condition itself (§1, and the companion note), must be the grade-2 space, P . So $[X, X]$ necessarily survives, nonzero, whenever P is the closed line.

Verification. Tested directly and exhaustively across all 56 valid partitions: $\{X, X\} \neq 0$ in exactly the 28 cases where P is the closed Fano line, and $\{X, X\} = 0$ in exactly the 28 cases where X is — without a single exception. The claim is not a pattern observed in examples; it holds with full generality across the entire structurally valid family.

4. Why the asymmetry is not arbitrary

The condition is asymmetric between X and P for a specific, physical reason, not merely a combinatorial one. $\{X, X\} \propto P$ is the non-associative content that defines R-flux — it is the bracket whose non-vanishing the entire construction exists to produce. $\{P, P\} = 0$, by contrast, is not a loss but the expected, required form of the standard R-flux algebra (momenta commute). Consequently:

- Closure on the **momentum** side removes a bracket ($\{P, P\}$) that was supposed to vanish anyway — no physical content is lost.
- Closure on the **coordinate** side removes the *only* bracket ($\{X, X\}$) carrying the algebra's defining feature — the result is R-flux in name only, with the flux itself absent.

This is why the two orbits, despite being related by an exact automorphism of the underlying octonion algebra, are not physically interchangeable: the automorphism relating them does not commute with which role — coordinate or momentum — the physical construction assigns to each subspace.

5. Consequence for the earlier finding

Restated precisely: of the 56 structurally valid, automorphism-related contractions in this family, **28 (all of Orbit B) yield the genuine non-associative R-flux algebra**, including Günaydin-Lüst-Malek's own construction as one instance; **28 (all of Orbit A) yield an ordinary associative Heisenberg algebra**, satisfying Malcev only vacuously. Malcev-preservation across the full family, reported in the companion note, remains correct as stated — but the physically meaningful genericity result is narrower and sharper than "56 of 140": it is **28 of 140 give genuine algebraic R-flux**, and this specific 28 is picked out by one exact, provable, mechanistically-explained condition, not by a further unexplained coincidence. The verdict on Orbit A is completed, rather than left as a dismissal, in a companion paper (*The Second Contraction*): the associative Heisenberg algebra Orbit A produces is exactly the endpoint of a second Inönü-Wigner contraction applied to Orbit B — the two orbits are two stations of one tower, the non-associative stratum and the associative, quantum-mechanical one.

References

- Companion note: *Malcev-Preservation Is Generic, Not Exceptional: All Structurally Valid Contractions of the Octonions in the Günaydin-Lüst-Malek Family Preserve the Malcev Identity.*
- Companion paper: *The Second Contraction: From Algebraic R-Flux to the Heisenberg Algebra, the Dilation Symmetry It Creates, and the $SO(3)$ Remnant of G_2 .*
- Günaydin, M., Lüster, D. & Malek, E., "Non-associativity in non-geometric string and M-theory backgrounds, the algebra of octonions, and missing momentum modes,"

arXiv:1607.06474.

- Weimar-Woods, E., "Contractions, generalized Inönü-Wigner contractions and deformations of finite-dimensional Lie algebras," *Rev. Math. Phys.* 12 (2000), 1505–1529.

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